- *Cybersecurity and encrypted control systems*
  Given the increasing cyber threats, these drones feature multi-layered cybersecurity protocols, AI-driven anomaly detection, and quantum-resistant encryption to safeguard communication, data integrity, and mission command from hostile interference.
- *Sustainable and efficient power systems*
  Integrating hybrid-electric engines, solar-assisted systems, and advanced energy management technologies, fourth-generation drones offer greater fuel efficiency, operational range, and environmental sustainability compared to earlier UAVs.

5.2 Historical development of fourth-generation drones

The evolution of fourth-generation drones represents a culmination of decades of technological advancement in automation, artificial intelligence, stealth design, and network-centric warfare. Emerging in the early 2010s, this generation evolved from the foundation laid by third-generation UAVs such as the RQ-4 Global Hawk, MQ-9 Reaper, and X-47B, which introduced long-endurance flight, limited autonomy, and combat integration. The defining leap of fourth-generation systems came with the application of AI-driven autonomy, swarm coordination, and digital battlefield integration, transforming drones from remotely piloted machines into self-governing, cognitive aerial entities.

The early roots of fourth-generation drone development can be traced to post-2010 defense modernization programs initiated by leading aerospace nations. The United States, through initiatives such as DARPA’s Skyborg, the Loyal Wingman Program, and the Next Generation Air Dominance (NGAD) project, pioneered AI-enabled drones capable of learning, adapting, and coordinating with manned aircraft. Similarly, Europe’s Future Combat Air System (FCAS), a collaboration between Dassault Aviation and